Technical University of Denmark

SPRING, 2025
Campus: Lyngby

MODEL-BASED MACHINE LEARNING

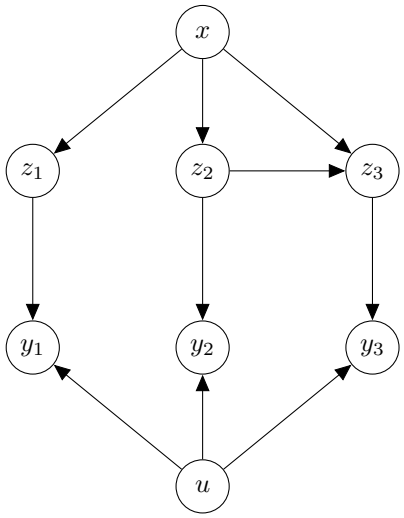
Mini-test 1

(Time allowed: 60 MINUTES)

Name: Kristin Böðvarsdóttir
Student ID: 253093

Part 1

Consider the following probabilistic graphical model (PGM):



1. (10%) Write the factorization of the joint distribution according to the PGM described above.

$$p(u)\,p(x)\,p(z_1|x)\,p(z_2|x)\,p(z_3|x,z_2)\,p(y_1|z_1,u)\,p(y_2|z_2,u)\,p(y_3|z_3,u)$$

2. (20%) Which of the following expressions correspond to valid expressions for $p(z_3|z_2,u)$ (more than option may the correct):

(a) $\frac{\sum_x \sum_{z_1} \sum_{y_1} \sum_{y_2} p(y,x,z_1,z_2,z_3,y_1,y_2,y_3)}{\sum_x \sum_{z_1} \sum_{z_3} \sum_{y_1} \sum_{y_2} p(y,x,z_1,z_2,z_3,y_1,y_2,y_3)}$

(b) $\sum_{y_1} p(z_3|z_2,u,y_1)$

(c) $\frac{\sum_x p(x)p(z_2|x)p(z_3|z_2,x)}{\sum_x p(x)p(z_2|x)}$ ←

(d) $\sum_{y_2} p(z_2|x)p(z_3|z_2,x)p(y_2|z_2,u)$

$$p(z_3|z_2,u) = \frac{p(z_3,z_2,u)}{p(z_2,u)}$$

$$= \frac{\sum_x p(z_3,z_2,u,x)}{\sum_x p(z_2,u,x)}$$

$$= \frac{\sum_x p(x)\,p(z_2|x)\,p(z_3|x,z_2)\,\cancel{p(u)}}{\sum_x p(z_2|x)\,p(x)\,\cancel{p(u)}}$$

$p(z_3|z_2,u)$

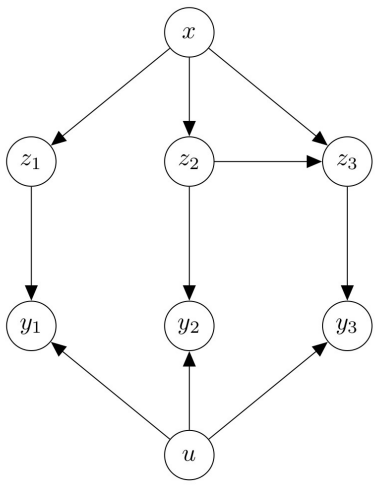
Skoða og setja á blað

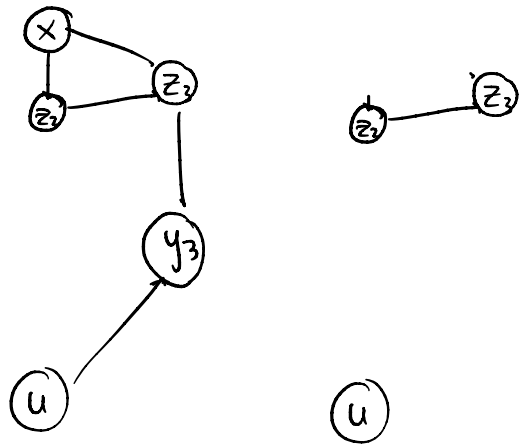
3. (15%) Classify the following statement as "True", "False", or "Cannot prove":

(a) $u \perp\!\!\!\perp z_3 | \{y_3, x\}$ True

(b) $x \perp\!\!\!\perp y_1 | \{z_1, y_2, y_3\}$ Cannot prove

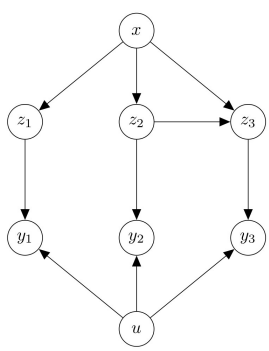
a)

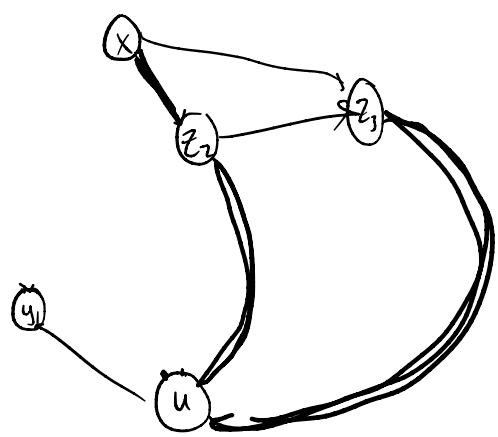




u and z_3 are disconnected

b)





x and y_1 are NOT disconnected

so cannot prove

Part 2

Consider the following generative process that models observed count data $\{y_{i,n}\}_{n=1}^{N_i}$ for each individual i using a hierarchical modelling approach:

1. Draw $\mu \sim \mathcal{N}(0, 1)$
2. Draw $v \sim$ Inverse-Gamma$(1, 1)$
3. For each individual $i \in \{1, \ldots, I\}$:
   (a) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mu, v)$ $N(\mu, v \mid x)$ μ
   (b) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:
      i. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

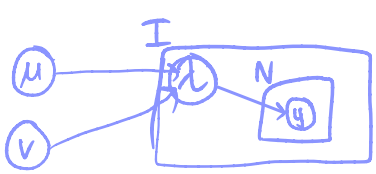
x

Make sure that you understand it well.

1. (15%) Write the factorization of the joint distribution according to the model described above.

$$P(\mu)\,p(v) \prod_{i=1}^{I} \left(P(\lambda_i \mid \mu, v) \prod_{n=1}^{N} P(y_{i,n} \mid \lambda_i) \right)$$

2. (15%) Draw the PGM representation corresponding to the generative process described above.



3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

(a) $p(u,\ \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(u, v, \boldsymbol{\lambda}, \mathbf{y})$

(b) $p(\boldsymbol{\lambda}|\ \mathbf{y}, u, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$

(c) $p(u|\ \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$

(d) $p(u|\ \lambda) = \frac{\int_v p(u)p(v)p(\boldsymbol{\lambda}|u,v)}{\int_v \int_{\boldsymbol{\lambda}} p(u)p(v)p(\boldsymbol{\lambda}|u,v)}$

4. (15%) Suppose now that you also have access to covariates $\mathbf{x}_i$ that describe characteristics of the each individual i that could help explain his/her latent log-rate λ_i. You may assume that the relationship between the mean of λ_i and $\mathbf{x}_i$ is linear in parameters. Which of the following generative processes corresponds to this enhanced model?

Option 1:

(a) Draw $v \sim$ Inverse-Gamma$(1, 1)$

(b) For each individual $i \in \{1, \ldots, I\}$:

  (i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mathbf{x}_i, v)$

  (ii) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:

    A. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

útilokunaradfest?

Option 2:

(a) Draw $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma\mathbf{I})$

rétt

(b) Draw $v \sim$ Inverse-Gamma$(1, 1)$

(c) For each individual $i \in \{1, \ldots, I\}$:

why ?

  (i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T\mathbf{x}_i, v)$

  (ii) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:

    A. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

Option 3:

(a) Draw $v \sim$ Inverse-Gamma$(1, 1)$

(b) For each individual $i \in \{1, \ldots, I\}$:

  (i) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:

    A. Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mathbf{x}_i, v)$

    B. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

Option 4:

(a) Draw $v \sim$ Inverse-Gamma$(1, 1)$

(b) For each individual $i \in \{1, \ldots, I\}$:

  (i) For each individual characteristic d:

    A. Draw $\beta_d \sim \mathcal{N}(0, \sigma^2)$

  (ii) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T\mathbf{x}_i, v)$

  (iii) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:

    A. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

individual β wouldn't make sense because how would we learn β if it's different for each individual ?

Part 2

Consider the following generative process that models observed count data $\{y_{i,n}\}_{n=1}^{N_i}$ for each individual i using a hierarchical modelling approach:

1. Draw $\mu \sim \mathcal{N}(0, 1)$ continuous
2. Draw $v \sim$ Inverse-Gamma$(1, 1)$
3. For each individual $i \in \{1, \ldots, I\}$:
   (a) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mu, v)$ $N(\mu, v \mid x)$ μ
   (b) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:
      i. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$

Make sure that you understand it well.

1. (15%) Write the factorization of the joint distribution according to the model described above.

$$P(\mu)\,P(v) \prod_{i=1}^{I} \left(P(\lambda_i \mid \mu, v) \prod_{n=1}^{N} P(y_{i,n} \mid \lambda_i) \right)$$

2. (15%) Draw the PGM representation corresponding to the generative process described above.

3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

(a) $p(u, \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(u, v, \boldsymbol{\lambda}, \mathbf{y})$

(b) $p(\boldsymbol{\lambda} \mid \mathbf{y}, u, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$ –

(c) $p(u \mid \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$ –

(d) $p(u \mid \lambda) = \frac{\int_v p(u)p(v)p(\boldsymbol{\lambda}|u,v)}{\int_v \int_{\boldsymbol{\lambda}} p(u)p(v)p(\boldsymbol{\lambda}|u,v)}$

y - discrete

v - continuous

$$= \frac{P(u, \lambda)}{P(\lambda)} = \frac{\int_v p(u, \lambda, v)}{\int_v \int_u p(u, v, \lambda)} =$$

a)

$$p(u, \vec{\lambda}) = p(u \mid \vec{\lambda})\, p(\vec{\lambda})$$

$$\int_v \sum_y p(u, v, \vec{\lambda}, \vec{y})$$

b)

$$p(\vec{\lambda} \mid \vec{y}, u, v) = p($$

3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

(a) $p(u,\ \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(u, v, \boldsymbol{\lambda}, \mathbf{y})$ T

(b) $p(\boldsymbol{\lambda} |\ \mathbf{y}, u, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$ T

(c) $p(u |\ \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(u, v, \boldsymbol{\lambda}, \mathbf{y})$ T

(d) $p(u |\ \lambda) = \frac{\int_v p(u)p(v)p(\boldsymbol{\lambda}|u,v)}{\int_u \int_v p(u)p(v)p(\boldsymbol{\lambda}|u,v)}$ F

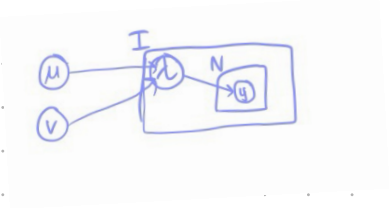
1. (15%) Write the factorization of the joint distribution according to the model described above.

$$p(\mu)p(v)\prod_{i=1}^{I}\left(p(\lambda_i|\mu,v)\prod_{n=1}^{N}p(y_{i,n}|\lambda_i)\right)$$

a) $p(u,\lambda) = \int_v \int_y p(u,v,\lambda,y)$

↑ y is discreet

b) $p(\lambda | y,u,v) \propto p(u,v,\lambda,y)$ T

c) $p(u | y,\lambda,v) \propto p(u,v,\lambda,y)$ T

d) $p(u|\lambda) = \frac{p(u,\lambda)}{p(\lambda)} = \frac{\int_v p(u)p(v)p(\lambda|u,v)}{\int_u \int_v p(u)p(v)p(\lambda,u,v)}$